EventShock

AI-Powered Financial Risk Intelligence & Portfolio Stress Testing

From fragmented financial information to **auditable portfolio risk insights.**

INFO
News, social posts, filings

→

SIGNAL
Event, severity, evidence

→

EXPOSURE
Portfolio stress result



▷ **Real EventShock Desk**, frozen replay, synthetic portfolio

S&P Global & Crisil Campus Hackathon 2026

Presented by **Aniket Mandal** | IIT Guwahati

github.com/Aniket03052006/IITG-AniketMandal-hackathon

Demo video: youtu.be/4Tynw6Z3xTM

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01 PROBLEM STATEMENT

Financial Risk Begins as Information. The Challenge Is Connecting It to Exposure.

Analysts lack **a single, reproducible path** from an external event to the portfolio positions it could affect.

SOURCES 01

Fragmented signals

News, social posts and filings arrive through **different channels**.

VOLUME 02

Information overload

Repeated reports and **noisy tone** complicate triage and hide what matters.

MAPPING 03

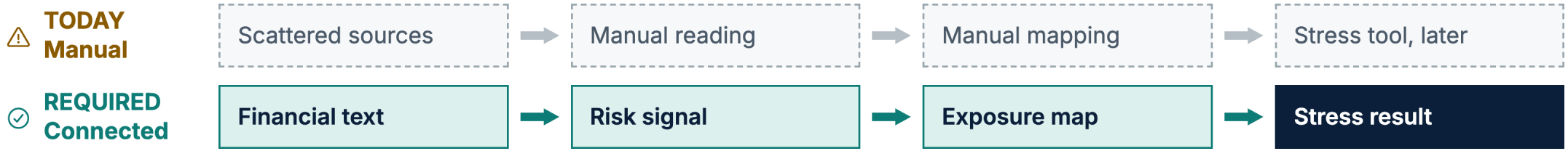
No exposure context

A headline does not say **which issuers, loans, bonds or derivatives** are hit.

WORKFLOW 04

Separate stress tools

Going from event to scenario is a slow, **manual hand-off** between tools.



⚠ **The cost:** delayed escalation, inconsistent treatment of similar events, and thin audit trails. A fast signal nobody can explain is unusable in a review; a thorough but slow process misses the window in which escalation helps.

Built for the Analyst Who Must Explain What an Event Means for the Portfolio

EventShock supports the analyst's decision; it **never changes positions or executes trades**.

PRIMARY PERSONA

Portfolio Risk Analyst

Monitors events, judges **materiality**, finds **exposed positions**.

KEY DECISION

"What needs review, and which exposures should we investigate?"

ALSO SERVES

Market and credit risk, risk managers who sign off, treasury

Needs: speed and a defensible number

TASK	FRICTION TODAY	EVENTSHOCK RESPONSE
Monitor information	Too many sources and repeated reports	→ **Unified ingestion** and de-duplication
Assess significance	Negative tone mistaken for materiality	→ **Separate** sentiment, event class and severity
Identify affected assets	Manual event-to-exposure mapping	→ Explicit scope and **exposure mapping**
Quantify the scenario	Separate, delayed stress workflow	→ Gated scenario and **deterministic engine**
Defend the result	Reasoning and assumptions hard to trace	→ **Evidence, provenance** and reproducible runs

EventShock prepares the evidence; the analyst makes the call.

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One Engine. Three Risk Readings. One Traceable Path to Portfolio Stress.

Severity is a **triage scale, not a loss probability**. Shock sizes are **assumptions, not forecasts**.

STEP 1
Ingest

News, posts, filings, pasted text; **one engine**

STEP 2
Clean & resolve

Validate, **de-duplicate**, resolve issuer

STEP 3
Three separate readings, never blended

Sentiment	Event class	Severity
-1 to +1	8 classes	1-10 scorecard
How framed?	*What happened?*	*How big?*

STEP 4
Evidence

Sources, quotes, **model hashes**; observed vs assumed

STEP 5
Map exposures

Event → risk factors → **issuers, sectors, positions**; every link has a rule, shocks bind by scope

Unreached exposure listed, never treated as safe

STEP 6
Run the stress test

$860.1M before

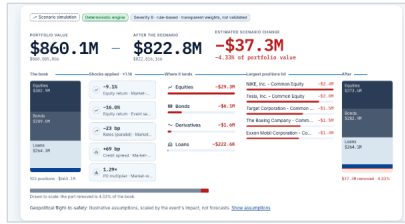
$822.8M after

Deterministic, -4.33%

Largest hit: **equities -$29.3M**

Run ID = hash of its inputs

STEP 7
Support review



Before/after, **top contributors**, unreached exposure, **audit trail**; every number opens to its inputs

From Unstructured Text to an Auditable Financial Risk Workflow

3,455 replay input records → **2,330** resulting signals → **27** reach severity ≥ 7 | **9-11 ms** stress-engine run | **2,368 + 782** backend + frontend tests

Funnel: **244 quarantined**, **184 exact duplicates folded** (separate outcomes). Models remain **experimental**.

Intelligence layer

- **Sentiment** score, -1 to +1
- 8-class **event** taxonomy
- Four-factor **severity** score
- **Abstains** below P 0.5
- Tone kept out of stress maths

Evidence layer

- **Provenance** and corroboration
- Model IDs + **artifact hashes**
- Duplicates folded
- Event→exposure links **traced**
- What-was-known-when timeline

Portfolio Stress Lab

- **103-position** synthetic book
- Equity/bond/loan/derivative
- **Loss attribution** by issuer
- **Unreached exposure** flagged
- **Reverse stress** testing

Product & engineering

- **FastAPI**, 93 operations
- **React 19**, Vite, TypeScript
- **Hash-pinned** replay
- Module A: index rebalancer
- Live feed off by default

WORKED EXAMPLE: GEOPOLITICAL

$860.1M → $822.8M

-$37.3M, or -4.33% of book value



Bonds -6.1, Derivatives -1.6, Loans -0.2 ($M)

84.5% of the book reached by the scenario

Top single position: **NIKE equity -$2.4M**

*Illustrative geopolitical stress scenario applied to a **synthetic** wholesale portfolio; shock sizes are **assumptions, not forecasts**.*

A Modular Architecture Built for Reproducibility

Data flows top to bottom. Sentiment, event class and severity are **separate components with separate outputs**.

1 Sources	**GDELT** news	Finance posts	**SEC EDGAR** 8-K	Pasted text	Prices, FOMC dates	Public, open data
2 Ingestion	Provider adapters	Validate & **quarantine**	Entity resolution	**De-duplication**		Python pipeline
3 Models	**Sentiment:** TF-IDF + logistic regression	**Event:** TF-IDF + E5-large vectors → LR				scikit-learn, **experimental**
	Impact ref.: additive gradient boosting	**Severity:** four-factor scorecard				
4 Risk logic	**Provenance**	Scope → factor mapping	Gate: **severity ≥ 7**	Labelled assumptions		Rules, not learned
5 Modules	**Module B: deterministic stress engine**	Module A: sentiment index rebalancer				Pure Python
6 App layer	**FastAPI**, OpenAPI	**React 19**, Vite, TypeScript	Files + in-memory repositories			Single process

Cross-cutting: fail loudly on modified artifacts, fixed replay clock, pinned digests

Deterministic maths
Code computes every stress number, **never an LLM**.

Evidence first
Every output keeps its **provenance** and inputs.

Controlled uncertainty
Missing or uncertain data is an **explicit state**.

Reproducibility
Frozen inputs and **SHA-256 hashes** repeat runs.

A Practical Bridge Between Financial Intelligence and Portfolio Risk Decisions

For S&P Global and Crisil: **a student prototype of the method**, not a claim to a better product.

WHO BENEFITS → DECISION

Portfolio & market risk

Find events needing review; **map factors to holdings**; run scenarios.

Credit risk

Surface issuer events and link them to **borrower exposures**, PD and spread shocks.

Risk governance

Check **evidence, assumptions** and model IDs before escalating.

Humans decide. EventShock never acts on its own.

PROPOSED OUTCOMES, STILL TO MEASURE

Intended benefit	Measurement required
Faster event triage	Publication → risk classification time
Less repeated review	Duplicate-grouping precision and recall
Better exposure mapping	Coverage vs analyst-labelled holdings
Faster stress analysis	Event → reproducible result time
Stronger auditability	Share of signals with full provenance

Measured today: a **9-11 ms stress run**. That is machine time, not analyst time saved.

BUSINESS MODEL

Proposed, not validated

Users: risk analysts, risk managers

Economic buyer: Head of Risk / CRO office, after model-risk and data sign-off

Possible revenue model

- Enterprise annual licence
- Deployment & integration services
- Optional specialised modules
- Licensed data & position-feed integration

No customers, pricing or market size claimed.

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Built to Demonstrate the Method. Designed with a Clear Path to Validation.

Reproducible decision support today; institutional reliance only after independent validation and production controls.

Model risk

- Event labels **AI-annotated**
- Impact skill ≈ **regime**
- Severity **not calibrated**
- Sentiment weak off-domain

Mitigation: labelled experimental; validate on human-reviewed data

Data risk

- Live feeds **rate-limited**
- Partial entity coverage
- Production needs **licences**
- Live poll not yet observed

Mitigation: frozen replay, quarantine, provenance, missing-data states

Financial-model risk

- Shock sizes **unvalidated**
- Thresholds are judgement
- **Synthetic** positions
- No curve or liquidity factor

Mitigation: labelled assumptions, unreached positions shown, expert review

Production risk

- **Single-process** prototype
- No auth or durable state
- No real position feed
- No drift monitoring yet

Mitigation: security, storage, access control and monitoring first

STAGE 1
Validate
Human labels, calibration, placebo tests

→

STAGE 2
Integrate
Licensed data, position feeds, risk systems

→

STAGE 3
Harden
Auth, durable storage, monitoring

→

STAGE 4
Pilot
Run beside analysts; scale on evidence

WHY IT WAS FEASIBLE

- Modular **Python** pipeline
- **CPU-only** classical ML
- Deterministic stress maths
- **2,368 + 782** tests passing
- Public, replayable data

Thank You

EventShock: From Financial Events to Portfolio Risk Insights

Turning **information** into **evidence**. Evidence into **exposure**. Exposure into **informed decisions**.

Information
News, posts, filings and pasted text

→

Evidence
Sources, quotes, model and data hashes

→

Exposure
Issuers, sectors, positions, not-reached

→

Decision
Made by the analyst, never the system

Built end to end and tested: 2,368 backend and 782 frontend tests

Every number traceable to its inputs, model and scenario

Limits stated: models experimental, shocks are assumptions

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